

4(a)(i)	A – <u>metaphase</u> ; B – <u>anaphase</u> ;	2
4(a)(ii)	any three from: <i>prophase</i> 1 form spindle, apparatus / fibres ; 2 attached to, centromere / kinetochore ; A remain attached to centromere during, prophase / metaphase / anaphase <i>(late) prophase</i> 3 move / AW, chromosomes / sister chromatids, to, (spindle) equator / metaphase plate or <i>during metaphase / until anaphase</i> maintain chromosomes at (spindle) equator ; A sister chromatids R chromatids unqualified <i>anaphase</i> 4 disassembly / shortening, of microtubules / spindle fibres ; 5 centromeres divide or chromatids move to (opposite), poles ; AW R sides 6 daughter nuclei receive, one chromatid of each chromosome ; A daughter nuclei receive a full complement of chromosomes / AW 7 AVP ; e.g. pole to pole spindle fibres lengthen to extend the cell	3

Question	Answer	Marks
4(a)(iii)	I telophase any three from: 1 nuclear envelope (re)forms around each group of chromosomes ; R nuclear membrane 2 uncoiling / decondensing / AW, of chromosomes ; A ref. to chromatin 3 nucleolus / nucleoli (in each nucleus), reform(s) / reappears / visible ; 4 organelles distributed between <u>two halves of cell</u> ; 5 cytokinesis / division of cytoplasm ; A cytoplasm splits 6 cleavage furrow forms ; A described e.g. 'pinching in' of (cell surface) membrane A equator of cell pinches in / cell membrane constricts R cell plate 7 AVP ; e.g. ref to ring of microfilaments / contractile ring, draws membrane closer together / AW I microtubules	3

Question	Answer	Marks
4(b)	any three from: 1 <u>semi-conservative replication</u> of DNA ; 2 <i>idea that</i> each new <u>chromatid</u> is identical to old one / sister chromatids are identical or each chromosome has two identical DNA molecules ; 3 alignment of chromosomes on the, (spindle) equator / metaphase plate ; 4 chromatids separate so each cell receives a (sister / identical) chromatid of each chromosome ; 5 centromere(s) split / shortening of spindle fibres ; 6 sister / identical, chromatids move to (opposite), poles ; AW A chromosomes / daughter chromosomes 7 checkpoints occur within cell cycle ; 8 prevents cells entering next stage if there are errors ; 9 (so that) errors in replication are repaired (by DNA polymerase) ; 10 to avoid mutations in daughter cells ;	3